5. Given the following data, what will be the possible yield when the rainfall is 29.

Details	Rainfall	Production
Mean	25``	40 units per acre
Standard Deviation	3``	6 units per acre

Coefficient of correlation between rainfall and production is 0.8.

6. The following data relate to advertisement expenditure (in lakh of rupees) and their corresponding sales (in crores of rupees)

Advertisement expenditure	40	50	38	60	65	50	35
Sales	38	60	55	70	60	48	30

Estimate the sales corresponding to advertising expenditure of ₹ 30 lakh.

7. You are given the following data:

Details	X	Y
Arithmetic Mean	36	85
Standard Deviation	11	8

If the Correlation coefficient between X and Y is 0.66, then find (i) the two regression coefficients, (ii) the most likely value of Y when $X = 10$.

8. Find the equation of the regression line of Y on X, if the observations (X_i, Y_i) are the following (1,4) (2,8) (3,2) (4,12) (5, 10) (6, 14) (7, 16) (8, 6) (9, 18).

9. A survey was conducted to study the relationship between expenditure on accommodation (X) and expenditure on Food and Entertainment (Y) and the following results were obtained:

Details	Mean	SD
Expenditure on Accommodation (₹)	178	63.15
Expenditure on Food and Entertainment (₹)	47.8	22.98
Coefficient of Correlation	0.43	

Write down the regression equation and estimate the expenditure on Food and Entertainment, if the expenditure on accommodation is ₹ 200.

10. For 5 observations of pairs of (X, Y) of variables X and Y the following results are obtained. $\Sigma X = 15$, $\Sigma Y = 25$, $\Sigma X^2 = 55$, $\Sigma Y^2 = 135$, $\Sigma XY = 83$. Find the equation of the lines of regression and estimate the values of X and Y if $Y = 8$; $X = 12$.

11. The two regression lines were found to be $4X-5Y+33 = 0$ and $20X-9Y-107 = 0$. Find the mean values and coefficient of correlation between X and Y.

12. The equations of two lines of regression obtained in a correlation analysis are the following $2X = 8-3Y$ and $2Y = 5-X$. Obtain the value of the regression coefficients and correlation coefficient.

Exercise 9.3

Choose the correct answer

1. Example for positive correlation is
   (a) Income and expenditure
   (b) Price and demand
   (c) Repayment period and EMI
   (d) Weight and Income

2. If the values of two variables move in same direction then the correlation is said to be
   (a) Negative
   (b) Positive
   (c) Perfect positive
   (d) No correlation

IL8BC4

3. If the values of two variables move in opposite direction then the correlation is said to be

09_11th_BM-STAT_Ch-9-EM.indd 227 21-04-2020 12:28:40 PM